MUSYOKI MUTUA

Full-Stack Software Developer

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Location: Nairobi, Kenya

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Portfolio: msyoki.github.io/profile-portfolio



SUMMARY

Full-Stack Software Developer with 6+ years of experience building secure, scalable enterprise web applications with Python (Django, Django REST Framework, FastAPI) and React (JavaScript, TypeScript, Vite). Specialized in document management, digital signing, workflow automation and system integration, including extending M-Files through custom applications and APIs, and in M-Pesa (Safaricom Daraja) payment integration and reconciliation. Also builds offline-first mobile apps with React Native. Delivers end-to-end solutions, from requirements and architecture through deployment and user training, for organizations across East Africa and the Horn of Africa.

EXPERIENCE

Consultant (Full-Stack Developer)

Techedge Africa – Consulting

03/2020 - Present

- Design and build full-stack enterprise applications with React and Python (Django, FastAPI), including RESTful APIs, backend services and integrations that support complex business processes.
- Built the company's Digital Signing System (DSS) and contributed to the architecture of its multi-tenant Enterprise Document Management System (EDMS), replacing manual, paper-based approvals for client organizations (see Projects).
- Implemented and extended M-Files document management solutions across multiple organizations, improving document control, workflow efficiency and operational visibility.
- Developed custom web applications and system integrations for clients including the University of Nairobi and WAICARE, enabling seamless data exchange between business systems and document management platforms.
- Conducted international system architecture, deployment and user training engagements in Djibouti, supporting successful adoption of enterprise solutions.
- Delivered solutions for IGAD, University of Nairobi, CARGEN, EARE, Mhasibu SACCO, Banki Kuu SACCO, Java House Africa and WAICARE, working with cross-functional teams and client stakeholders from requirements gathering through deployment and post-launch support.

EDUCATION

Certificate in Software Development, Computer Software Technology/Technician

Moringa School

02/2020 - 11/2020

Bachelor of Commerce (B.Com.), Business Administration & Management Science

Strathmore University

2010 - 2014

SKILLS

Frontend

React | Next.js | Vite | TypeScript | JavaScript | Tailwind CSS | Material UI | Bootstrap

Mobile

React Native | Expo | Zustand | SQLite (offline-first)

Backend

Python | Django | Django REST Framework | FastAPI | Node.js

Databases

PostgreSQL | MySQL | Supabase | Drizzle ORM

APIs & Integration

M-Pesa Daraja | STK Push | PayPal | REST APIs | gRPC | JWT / SSO | LDAP

Tools

Git | GitHub Actions | Docker | Linux | Nginx | IIS | Playwright | Claude Code

Enterprise systems

M-Files | ERPNext | WordPress | WooCommerce

EXPERIENCE

Freelance Full-Stack Developer

2020 - Present

Custom Web Application Development (React, Django, FastAPI)

- Designed and developed custom web applications, dashboards and business automation tools using React, Django, FastAPI and REST APIs for clients across multiple industries.
- Built secure backend services, authentication systems and third-party integrations, including M-Pesa, PayPal and other external APIs.
- Delivered CMS-based websites and e-commerce stores with WordPress and WooCommerce where they best fit client requirements.
- Digitized manual business processes and integrated multiple systems into unified platforms, from requirements gathering through deployment and post-launch support.

ENTERPRISE PROJECTS

Digital Signing System (DSS) — Multi-Tenant E-Signature Platform

React, Vite, Material UI, pdf-lib, PDF.js, Django REST Framework, PostgreSQL, Twilio, Docker, Nginx

Techedge Africa

- Built an end-to-end e-signature platform: upload a document, place signature fields by drag and drop, set the signing order and route it to internal and external signers, or sign it yourself.
- Secured signing with JWT authentication and SMS one-time codes via Twilio, with a full audit trail of every view, signature, comment and void action, including the signer's IP address.
- Supports multiple companies on one platform, each with its own plan, branding, users, reusable workflows and SMTP email settings for notifications and reminders.
- Deployed with Docker and Nginx, and integrated with enterprise document management systems, sharing single sign-on with the EDMS.

Enterprise Document Management System (EDMS) — Confidential Enterprise Project

React, Vite, Django, JWT, REST APIs

Techedge Africa

- Developed a modern document management web application with React and Vite for document and metadata management, search, workflows and file handling.
- Built a centralized JWT authentication service in Django providing single sign-on and role-based permissions across the EDMS and DSS.
- Developed reusable React components, dynamic metadata forms, lookup controls and document preview, extending an established enterprise document management platform.
- Contributed to a reusable, multi-tenant platform designed for deployment and customization across multiple organizations.

OTHER WORK

- Daybook: offline bookkeeping app for a second-hand clothing shop (React Native, SQLite).
- Moirai: nonprofit website with M-Pesa and PayPal donations (React, TypeScript, Tailwind CSS, Netlify Functions).
- Counseling session booking app with Cal.com integration and Playwright end-to-end tests.
- Client websites for Steve Safaris, EMM Labs and DesignMotives (HTML/CSS, WordPress).

PROJECTS

Private repositories; code walkthroughs available on request.

Kodibox — Rent & Property Management for Kenya

Next.js 15, React 19, TypeScript, PostgreSQL 16, Drizzle ORM, Turborepo, Docker, GitHub Actions

In development

- Rent goes straight to the landlord's own M-Pesa paybill or bank account (C2B, STK Push), with automatic reconciliation and a review queue for unclear payments.
- Double-entry ledger with reversals instead of deletions; each landlord's data isolated by PostgreSQL row-level security.
- Built for Kenya's Data Protection Act 2019: consent records, encrypted ID and KRA PINs, audit logs, CI with CodeQL scanning.

ShulePlan — School Organizer App for Parents

React Native (Expo), TypeScript, SQLite, Supabase

Android app, pilot stage

- Each child's homework, exams, events and fees in one place, shared between both parents.
- Offline-first: reads from an on-device SQLite database and syncs to Supabase (Postgres with row-level security, Auth, Realtime).
- Records fee payments from a pasted M-Pesa SMS; term reports and reminders.

F8 — Business Card & Spend-Management App

React Native (Expo), React 19, TypeScript, Expo Router, Zustand, Zod, Figma, GitHub Actions

Fintech mobile prototype, co-developer

- Lead contributor (129 of 236 commits) to a Kenyan fintech prototype: onboarding, KYB/KYC requirements, phone and email verification, dashboard, card, payments and transactions screens.
- Built a design-token theme and shared UI component library from the Figma design system, with typed forms (react-hook-form, Zod) and Zustand state.
- Set up quality gates: strict TypeScript, linting, Conventional Commits and CI checks on pull requests.

Chama MGS — Savings Group Contributions

Django REST Framework, React, TypeScript, Daraja API, Docker

Personal project

- Confirms contributions through Daraja C2B with automatic matching and duplicate prevention.
- Allocates payments to the oldest month first, with partial payments, late fees and audited reversals; 65 automated tests.

Kenya Ebola Live Tracker

Python, GitHub Actions, GitHub Pages

Personal project

- Checks Ministry of Health, WHO and Africa CDC sources every 30 minutes; reports are verified by a person before publishing.